Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/19/2006
PI Name:	Koplik, Joel Ph.D.		
Project Title:	Molecular Dynamics of Fluid-Solid Systems		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10031	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	11/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2007 per PI (10/06) Note - grant renewed for 4 yrs/new NAG#		
Key Personnel Changes/Previous PI:	Koplik, Joel		
COI Name (Institution):	Banavar, Jayanth (Pennsylvania State University)		
Grant/Contract No.:	NNC04GA27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The numerical calculations are based on Molecular Dynamics, wherein one integrates the (classical) equations of motion for interacting molecules of various species, based on a prescribed interaction and specified thermodynamic operating conditions and mechanical forces. Within the framework of molecular dynamics, it is straightforward to impose a desired value of gravity or simulate the presence of g-jitter. The objective of this theoretical research is to examine the molecular behavior of fluids in confined geometries, to determine flow boundary conditions, and to investigate non-Newtonian rheology, using molecular dynamics calculations and the statistical mechanics of classical fluids. We consider temporal and spatial scales where continuum fluid mechanics provides little insight. Examples under investigation include (1) phase behavior and spreading of surfactant solutions, (2) solidification, evaporation, coalescence and rupture of non-Newtonian fluid drops and filaments, (3) boundary conditions for dilute gases and other interfaces (4) dynamics of a polymer molecule in solution, and (5) the physics of colloidal hard sphere systems. We intend to carry out a number of calculations involving coexisting liquid, vapor and solid phases under flow conditions, with two common features. First, the systems will be studied at the molecular scale to obtain otherwise unavailable information. Second, the problems to be studied deal with fundamental issues in the physics of fluids and interfaces that are of relevance to the microgravity program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research provides an improved understanding of the microscopic physics underlying macroscopic fluid pheneomena in several areas, which is expected to lead to better design and fabrication in future applications.
Task Progress:	1. Recent advances in fabrication techniques have spawned the field of nanofluidics where minute amounts of fluid are contained and controlled for practical use. Understanding the behavior of fluids sometimes requires information at the molecular scale, e.g., about the boundaryconditions. The continuum description itself may break down as in a rarified fluid in the Knudsen regime in which the mean free path is large. Computer simulations bridge molecular scale physics and the continuum behavior and are ideally suited to study nanosized systems. Understanding of liquid and gas flows in nanoscale devices is of great technological interest and requires a molecular scale analysis both because the boundary conditions determined by the fluid-solid interactions are all-important and because of dilution. Molecular dynamics simulations of dense and rarefied fluids comprising small chain molecules in chemically patterned nanochannels were carried out and predicted a novel switching from Poiseuille to plug flow along the channel. We also demonstrated behavior akin to the lotus effect for a nanodrop on a chemically patterned substrate. Our results show that one can control and exploit the behavior of fluids at the nanoscale using chemical patterning. 2. The conventional boundary conditions at the interface between two flowing liquids include the continuity of the velocity field. As in the liquid-solid case, continuity of the tangential component of velocity is not obvious, and we have undertaken molecular dynamics simulations of the Couette and Poiseuille flows of two-layered liquid systems with various molecular structures and interactions. In all cases, the average velocity is found to vary continuously across the interface. However, when the total liquid density in the interfacial region drops significantly compared to the bulk values, the tangential velocity varies very rapidly there, and would appear discontinuous at a coarse or continuum resolution. The value of this apparent slip varies linearly with the shear stress at the interface, with a constant of proportionality depending on the nature of both liquids but not the flow configuration. Thus, a version of the Navier boundary condition appears to apply in this situation. 3. Micro- and nano-scale fabrication techniques allow one to pattern a surface with distinct regions which may be either wetting or non-0wetting for a particular fluid. We have investigated the flow of a nano-scale incompressible ridge of low-volatility liquid along a ``chemical channel': a long, straight, and completely wetting stripe embedded in a planar substrate, and sandwiched between two extended less wetting solid regions. Molecular dynamics simulations, a simple long-wavelength approximation, and a full stability analysis based on the Stokes equations are used, and give qualitatively consistent results. While thinl iquid ridges are stable both statically and during flow, a (linear) pearling instability develops if the thickness of the ridge exceeds half of the width of the channel. In the flowing case periodic bulges propagate along the channel and subsequently merge due to nonlinear effects. However, the ridge does not break up even when the flow is unstable, and the qualitative behavior is unchanged even when the fluid can spill over onto a partially wetting exterior solid region.
Bibliography Type:	Description: (Last Updated: 12/21/2007)
Articles in Peer-reviewed Journals	Cieplak M, Koplik J, Banavar JR. "Nanoscale fluid flows in the vicinity of patterned surfaces." Phys Rev Lett. 2006 Mar 24;96(11):114502. Epub 2006 Mar 23. PMID: 16605828 , Mar-2006
Articles in Peer-reviewed Journals	Koplik J, Lo TS, Rauscher M, Dietrich S. "Pearling instability of nanoscale fluid flow confined to a chemical channel." Phys Fluids 2006 Mar;18(3):032104. , Mar-2006
Articles in Peer-reviewed Journals	Koplik J, Banavar JR. "Slip, immiscibility, and boundary conditions at the liquid-liquid interface." Phys Rev Lett. 2006 Feb 3;96(4):044505. Epub 2006 Feb 2. PMID: 16486831 , Feb-2006